

7.1 How Vector Graphics Work

The images you encounter nearly everywhere are called *Raster* images, or simply *rasters*. In plain words, raster images are a large collection of pixels—to draw the raster, you need to provide information about each of the pixels. Vector images (or “vectors” for convenience), on the other hand, use primitive shapes such as lines, curves and polygons rather than pixels.

Vector images are a smarter way of storing graphics information—let’s see how. To draw a red circle in the raster format, an image editing program will plot pixels on to the image one pixel at a time—something like “One red pixel here, one red pixel there,” and so on. Drawing the same circle as a vector image makes a lot more sense—it draws it the way you would draw it on paper, using the following information:

- Where the centre of the circle is located
- The radius of the circle
- The colour of the circle

Armed with this information and a few mathematical calculations, the image editing program can calculate the size of the image in pixels—because ultimately, everything that is displayed on your screen is going to be translated into pixels.

Obviously, this is a lot less information to store. Suppose you want the circle bigger. For the raster, you will need to add more and more pixels, each of which comes with a baggage of three bytes (one each for the Red, Green and Blue levels) to add to the file size. On the other hand, to change the size of the vector graphic, you’re just going to change the value of the radius—something that is going to make a marginal, if any, difference to the file size.

Resizing Vectors and Rasters

Let’s say that we saved the aforementioned circle image as a 640 x 480 bitmap file and now wanted to change its size to 1024 x 768.